Keyu Chen

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CurryChen77 | Google Scholar

Beijing, China

RESEARCH INTERESTS

My primary research focus is **reinforcement learning post-training (RL post-training)** for closed-loop policy optimization in autonomous driving. I study how reinforcement learning and interactive feedback improve the adaptability, robustness, and generalization of pre-trained policies. My broader interests span **Physical AI**, including learning-based decision-making and policy optimization for embodied agents in complex environments.

EDUCATION

- **Tsinghua University (THU)** *September 2023 - July 2028(Expected)*
  Ph.D. student at the School of Vehicle and Mobility. Beijing, China
  - Advisor: Prof. Sifa Zheng.

- **Nanjing University of Aeronautics and Astronautics (NUAA)** *September 2019 - July 2023*
  B.Eng. in Vehicle Engineering. Nanjing, China
  - GPA: 4.1/5 (Top 2%)

FIRST AUTHOR PUBLICATIONS

[NeurIPS 2026] **CRAFT: Counterfactual-to-Interactive Reinforcement Fine-Tuning for Driving Policies.**
Keyu Chen, Nanfei Ye, Yida Wang, Wenchao Sun, Danqi Zhao, Hao Cheng, Sifa Zheng.
Paper, Project Page, Code
CRAFT performs closed-loop RL post-training of pre-trained driving policies by combining dense counterfactual proxy supervision with residual correction from closed-loop interaction. Self-distillation stabilizes optimization, improving closed-loop driving performance across multiple policy families.

[ICRA 2026] ForSim: Stepwise Forward Simulation for Traffic Policy Fine-Tuning.
Keyu Chen, Wenchao Sun, Hao Cheng, Zheng Fu, Sifa Zheng.
Paper, Project Page, Code
ForSim achieves high-fidelity traffic simulation by combining stepwise closed-loop trajectory propagation with physical dynamics and interaction-aware agent updates.

[arXiv 2025] RIFT: Group-Relative RL Fine-Tuning for Realistic and Controllable Traffic Simulation.
Keyu Chen, Wenchao Sun, Hao Cheng, Sifa Zheng.
Paper, Project Page, Code, Stars **130+**
RIFT achieves realistic and controllable traffic simulation by combining IL pre-training in a data-driven simulator for realism with RL fine-tuning in a physics-based simulator for controllability.

[CoRL 2024] FREA: Feasibility-Guided Generation of Safety-Critical Scenarios with Reasonable Adversariality.
Keyu Chen, Yuheng Lei, Hao Cheng, Haoran Wu, Wenchao Sun, Sifa Zheng. (**Oral 4.3%**)
Paper, Project Page, Code, Stars **60+**
FREA incorporates feasibility as guidance to generate adversarial yet AV-feasible, safety-critical scenarios for autonomous driving.

[KBS 2023] IGT: Illumination-guided RGB-T object detection with transformers.
Keyu Chen, Jinqiang Liu, Han Zhang. (Bachelor's thesis)
Paper
IGT uses illumination intensity to guide the fusion process of multi-modality features, enabling the comprehensive utilization of cross-modality complementary information in object detection.

Co-Author Publications

[ECCV 2026] SparseDriveV2: Scoring is All You Need for End-to-End Autonomous Driving.
Wenchao Sun, Xuewu Lin, **Keyu Chen**, Zixiang Pei, Xiang Li, Yining Shi, Sifa Zheng.
Paper, Code

[RA-L 2026] DriveCamSim: Generalizable Camera Simulation via Explicit Camera Modeling for Autonomous Driving.
Wenchao Sun, Xuewu Lin, **Keyu Chen**, Zixiang Pei, Yining Shi, Chuang Zhang, Sifa Zheng.
Paper, Code

[TRC 2025] Emergency Index (EI): A two-dimensional surrogate safety measure considering vehicles' interaction depth.
Hao Cheng, Yanbo Jiang, Hailun Zhang, **Keyu Chen**, Heye Huang, Shaobing Xu, Jianqiang Wang, Sifa Zheng.
Paper, Code

Internships

- **Li Auto** *November 2025 - Present*
  Closed-loop RL post-training for autonomous driving. Beijing, China
- **Tsinghua Automotive Research Institute (Suzhou)** *July 2023 - August 2023*
  Safety-critical scenario generation for autonomous driving. Suzhou, China

Honors and Awards

- **Outstanding Graduate (Top 1%), NUAA** *June 2023*
- **Outstanding Bachelor Thesis Award (Top 5%), NUAA** *June 2023*
- **National Scholarship (Top 1%), NUAA** *December 2021*